

Summer Camp 2026 Week 4 Day 2

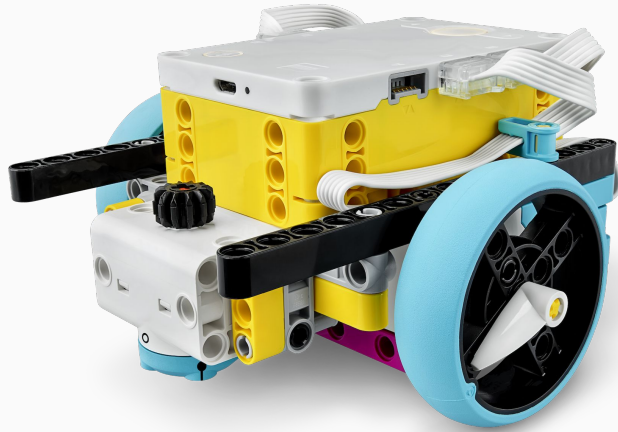
Ahmaad Jackson

Intro

- What is Lego Spike Prime?
 - How does it differ from Spike Essentials?
- What are motors?
 - How can you control motors in the Spike App?
- What is a robot?

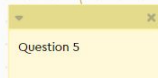
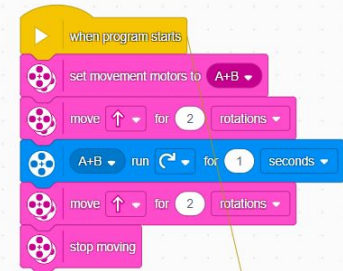
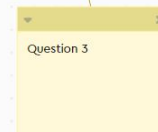
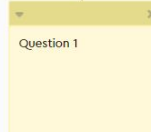
Build a 2 Wheel Base

- 1 Hub
- 2 Large Motors
- 2 Wheels



Spike Prime Intro

1. Use this code
2. Download & Run: Does the robot move forward?
3. Change 3 rotations to 5.
Does it go farther?
4. Add: Move forward 2 rotations" -> "Turn right 1 second" -> "Move forward 2 rotations."



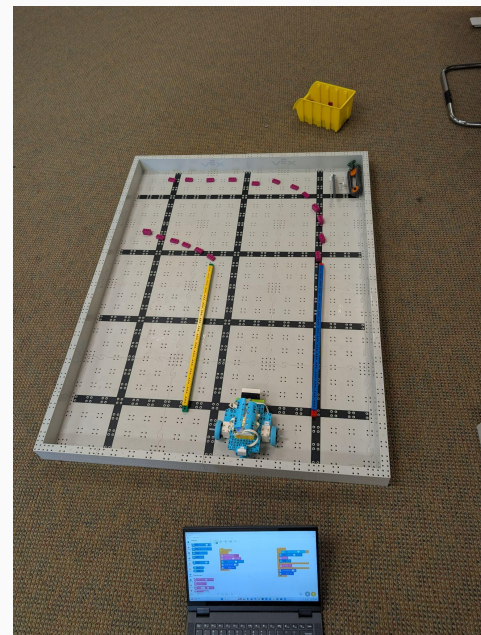
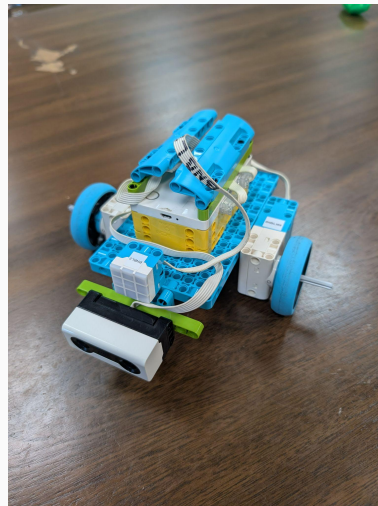
Break Time

- Let's take our break!
- We'll come back at 2:30

Mini-Robot

- The robot can now move & turn
 - Let's see if it can navigate a course
- Navigate a course
- Autonomously
- 3 Attempts
 - Adjust Robot after each attempt



[illegible]

Testing Details

- Date Tested: 07/21/2026
- Issue: LSP Mini Robot Challenge
- Environments: Build, Lego Spike Prime, Live Test Track
- Test Time: 1 Hour 15 Mins (2:30 PM to 3:45 PM)
- Description: Went over an introduction to Lego Spike Prime. Built a 2 wheel base for a robot, and then programmed it to test its movement around a simple track. Completed 4 attempts using the engineering design process to evaluate each test.

Test Script | Sample Test 1

Attempt	Description	Expected Result	Actual Result	Pass/Fail
1	Robot navigates the course. Straight + Left Curve. Use rotation for the motor programming.	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight & does not turn.	Fail
2	Robot navigates the course. Straight + Left Curve. Use rotation for the motor programming.	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, turns left when it hits object. Does not continue to move straight after left turn.	Fail
3	Robot navigates the course. Straight + Left Curve. Use distance for the motor programming. (Use Distance Sensor)	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, hits the outer curve, reverses, & faces left. Does not continue to move straight after left turn.	Fail
4	Robot navigates the course. Straight + Left Curve. Use distance for the motor programming. (Use Distance Sensor)	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, hits the outer curve, reverses, & faces left. Moves straight after left turn.	Pass (Adjust to fine-tune)

Test Script | Sample Test 2 (Me)

Attempt	Description	Expected Result	Actual Result	Pass/Fail
1	Robot navigates the course. Straight + Left Curve. Use rotation for the motor programming.	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight. Turns left, but not completely.	Fail
2	Robot navigates the course. Straight + Left Curve. Use rotation for the motor programming.	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, turns left through track & moves straight. But it hits 1 small piece of the track on the inner curve.	Pass (Adjust to fine-tune)
3	Robot navigates the course. Straight + Left Curve. Use rotation for the motor programming. (Widened the curve on the track)	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, turns left through track & moves straight.	Pass
4	Robot navigates the course. Straight + Left Curve. Use distance for the motor programming. (Use Distance Sensor)	Robot will move straight. Then turn left before it hits the edges of the curve.	Robot moves straight, sees the curve barriers, reverses, faces left, & moves straight. Repeats with each barrier.	Pass (Adjust to fine-tune)

Adjustments made

Improvements with Each Attempt

- Word Blocks
 - Rotations
 - Proper Turning (Using Motors)
 - Distance Calculation
 - Fixing Control Loops
- Adjust Track / Widen Track